

Rotter, T., Veronig, A.M., Temmer, M. et al. Real-Time Solar Wind Prediction Based on SDO/AIA Coronal Hole Data. Sol Phys 290, 1355–1370 (2015)

SDO/AIA-193 Å observations and Advanced Composition Explorer measurements during the years 2011 – 2013.

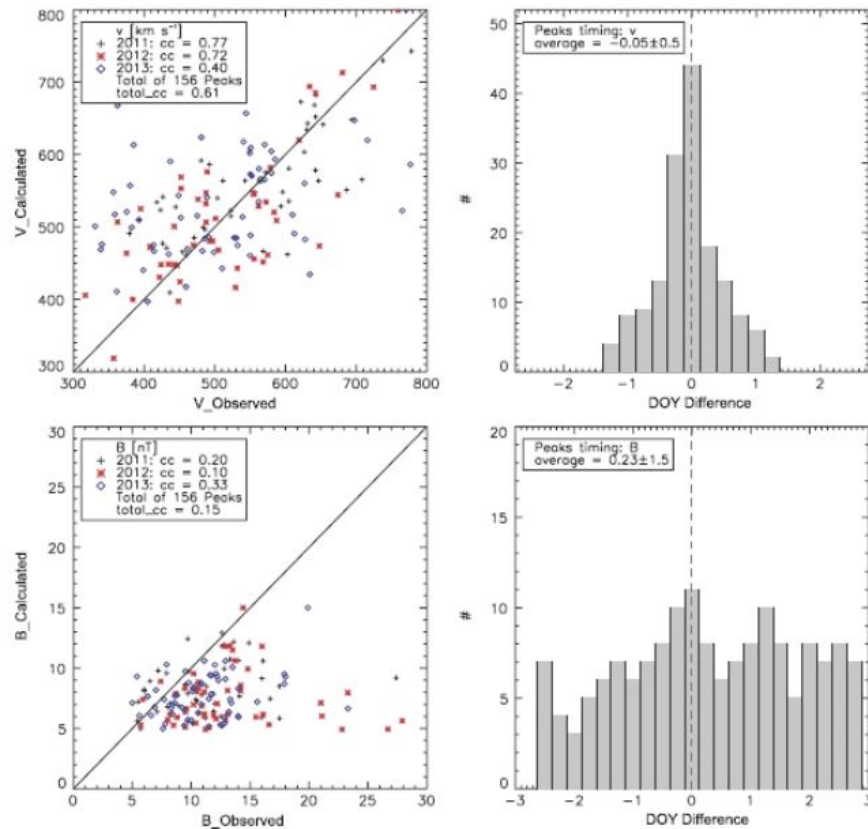
$$v(t+\tau) = c_0 + c_1 A(t)$$

These parameters are automatically updated by using a moving window of the duration of three Carrington Rotations

a lead time of four days

Solar wind data at 1 AU

The algorithm produces good predictions for the 156 solar wind high-speed streams peak amplitudes with correlation coefficients of $cc \approx 0.60$. For 80 % of the peaks, the predicted arrival matches the ACE in situ measurements within a time window of 0.5 days.



Left panels: Scatter plot of predicted versus measured peaks in the solar wind speed (top) and total magnetic field strengths (bottom). In total, 156 peaks were identified in the CH areas during the years 2011 – 2013. The related correlation coefficients are indicated in the insets (2011: plus sign; 2012: red stars; 2013: blue diamonds). The one-to-one correspondence is plotted as a black line. Right panels: Histograms of the difference of arrival times of measured and predicted solar wind v and B .

Reiss, M. A., M. Temmer, A. M. Veronig, L. Nikolic, S. Vennerstrom, F. Schöngassner, and S. J. Hofmeister (2016), Verification of high-speed solar wind stream forecasts using operational solar wind models. Space Weather

$$v_{sw}(R_s, \theta, \phi) = a_1 + \frac{a_2}{(1 + f_s)^{a_3}} \left[a_4 - a_5 \exp \left(- \left(\frac{\theta_b}{a_6} \right)^{a_7} \right) \right]^{a_8}, \quad (3)$$

where a_1 – a_8 are empirical numerical coefficients, $f_s = (|B(R_\odot)|R_\odot^2)/(|B(R_s)|R_s^2)$ is the flux tube expansion factor of an open magnetic field line, and θ_b is the angular separation between the coronal hole boundary and the open magnetic field line foot point at the photosphere. In this paper we use $a_1=250$, $a_2=875$, $a_3=0.2$, $a_4=1$, $a_5=0.8$, $a_6=2.6$, $a_7=1.25$, and $a_8=2.5$. From the source surface to 1 AU, the solar wind streams are propagated using a simple kinematic model [Arge and Pizzo, 2000].

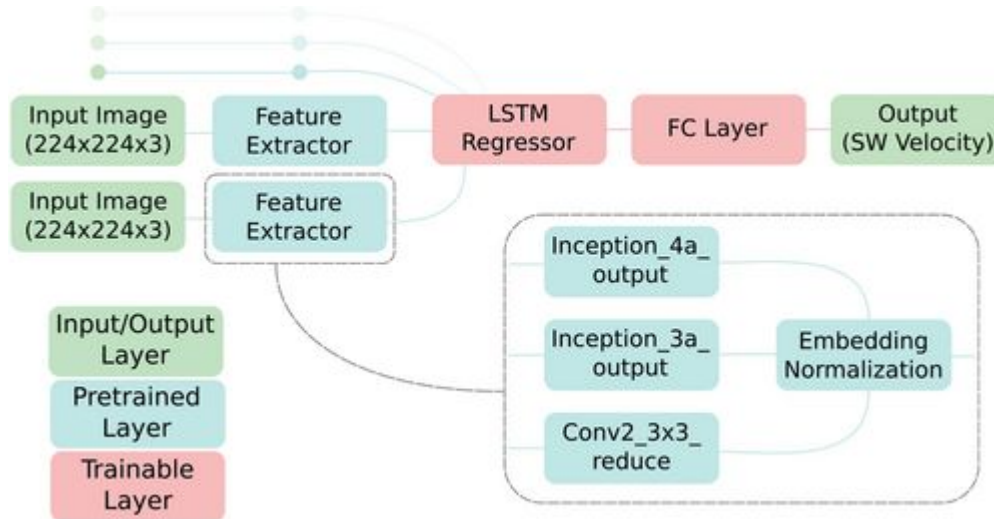
ME for the ESWF is 17.49 km/s, the RMSE is 108.2 km/s, and the correlation coefficient is 0.31

Upendran, V., Cheung, M. C. M., Hanasoge, S., & Krishnamurthi, G. (2020).
Solar wind prediction using deep learning. Space Weather, 18, e2020SW002478.

EUV images in 193 and 211 Å from Atmospheric Imaging Assembly (AIA) onboard SDO as input, with marked coronal holes and active regions.

Predict the solar wind speed present in the NASA OMNIWEB data set. (L1)

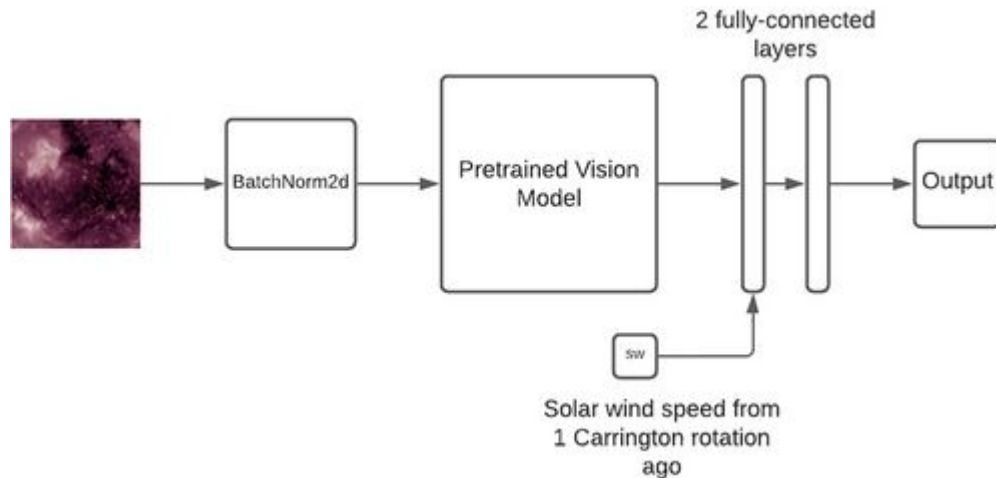
our WindNet may appear to not outperform existing models in terms of the metrics used



Brown, E. J. E., Svoboda, F., Meredith, N. P., Lane, N., & Horne, R. B. (2022). Attention-based machine vision models and techniques for solar wind speed forecasting using solar EUV images. Space Weather

EUV images at 211 Å

Solar wind speed from WIND and the Advanced Composition Explorer spacecraft, both positioned at the L1 point



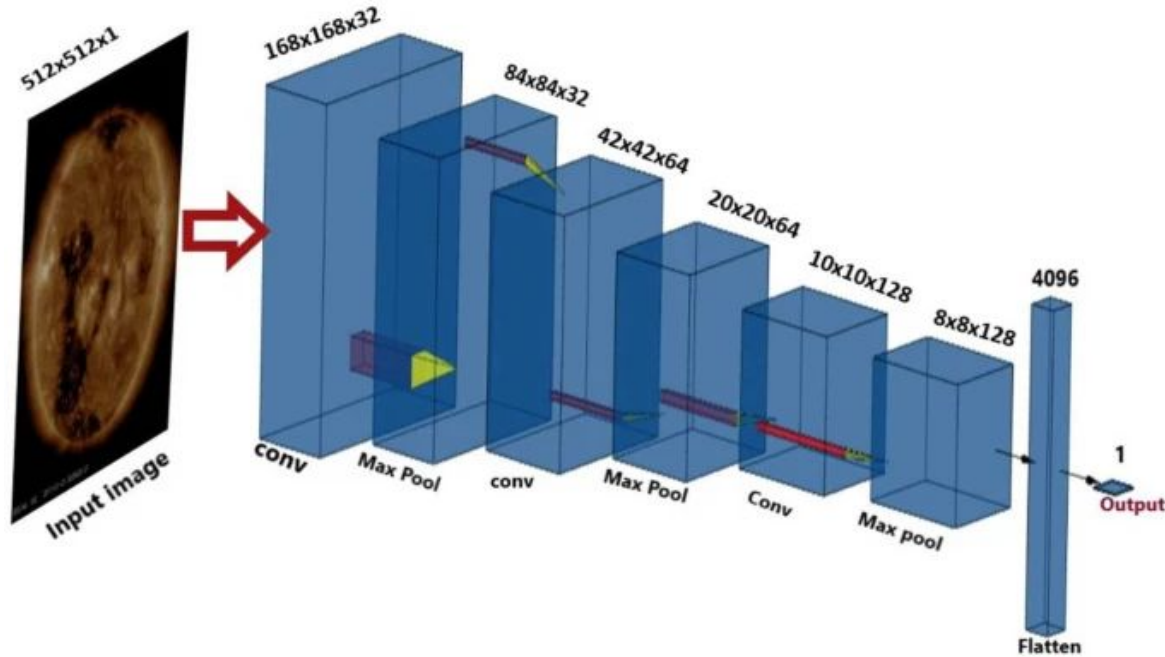
	RMSE	% Improvement	Correlation	% Improvement
Model				
Persistence (4 day)	127.59	-57.1%	0.080	-85.2%
Persistence (27 day)	100.86	-24.2%	0.426	-21.1%
Upendran et al. (2020)	81.21	-	0.54	-
Our models				
Solar InceptionNet v4	74.09	8.8%	0.609	12.7%
Solar DenseNet	73.92	9.0%	0.611	13.1%
Solar GoogleNet	73.71	9.2%	0.619	14.6%
Solar ResNet	73.52	9.5%	0.618	14.4%
Solar TNT	72.70	10.5%	0.629	16.5%
Solar Vision Transformer	72.66	10.5%	0.630	16.7%
Solar Swin Transformer	72.21	11.1%	0.634	17.4%

Raju, H., Das, S. CNN-Based Deep Learning Model for Solar Wind Forecasting. Sol Phys 296, 134 (2021)

Images from the *Atmospheric Imaging Assembly* (AIA) at 193 Å wavelength are used for training. Solar-wind speed is taken from the *Advanced Composition Explorer* (ACE) located at the Lagrangian L1 point.

RMSE 76.30 ± 1.87 km/s

correlation coefficient of 0.57 ± 0.02



Bernoux, G., Brunet, A., Buchlin, É., Janvier, M., & Sicard, A. (2022). Forecasting the geomagnetic activity several days in advance using neural networks driven by solar EUV imaging. Journal of Geophysical Research: Space Physics

Input: sequences of 10 images at 19.3 nm with 12 hr cadence

Predict with a mean and sigma for the daily max kp-index

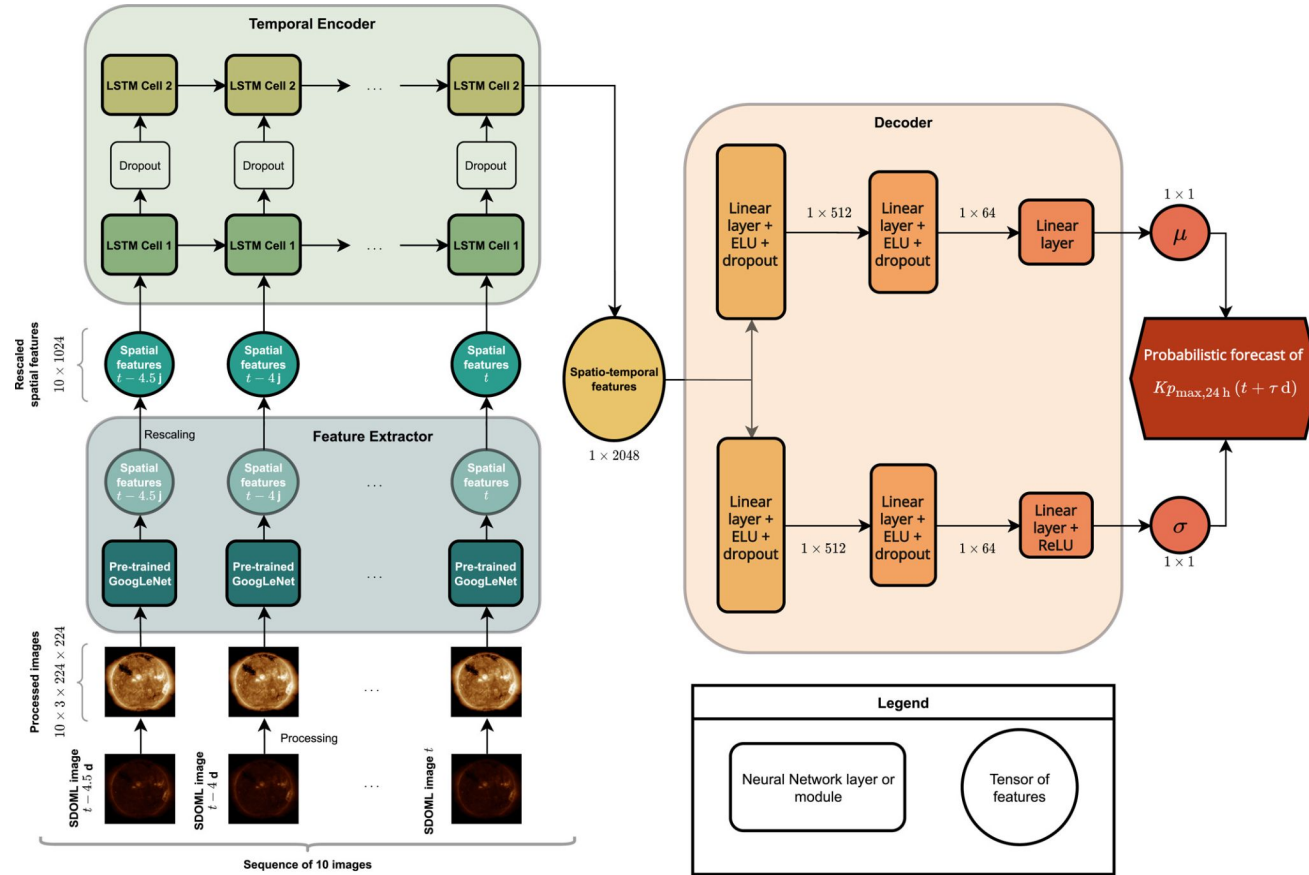
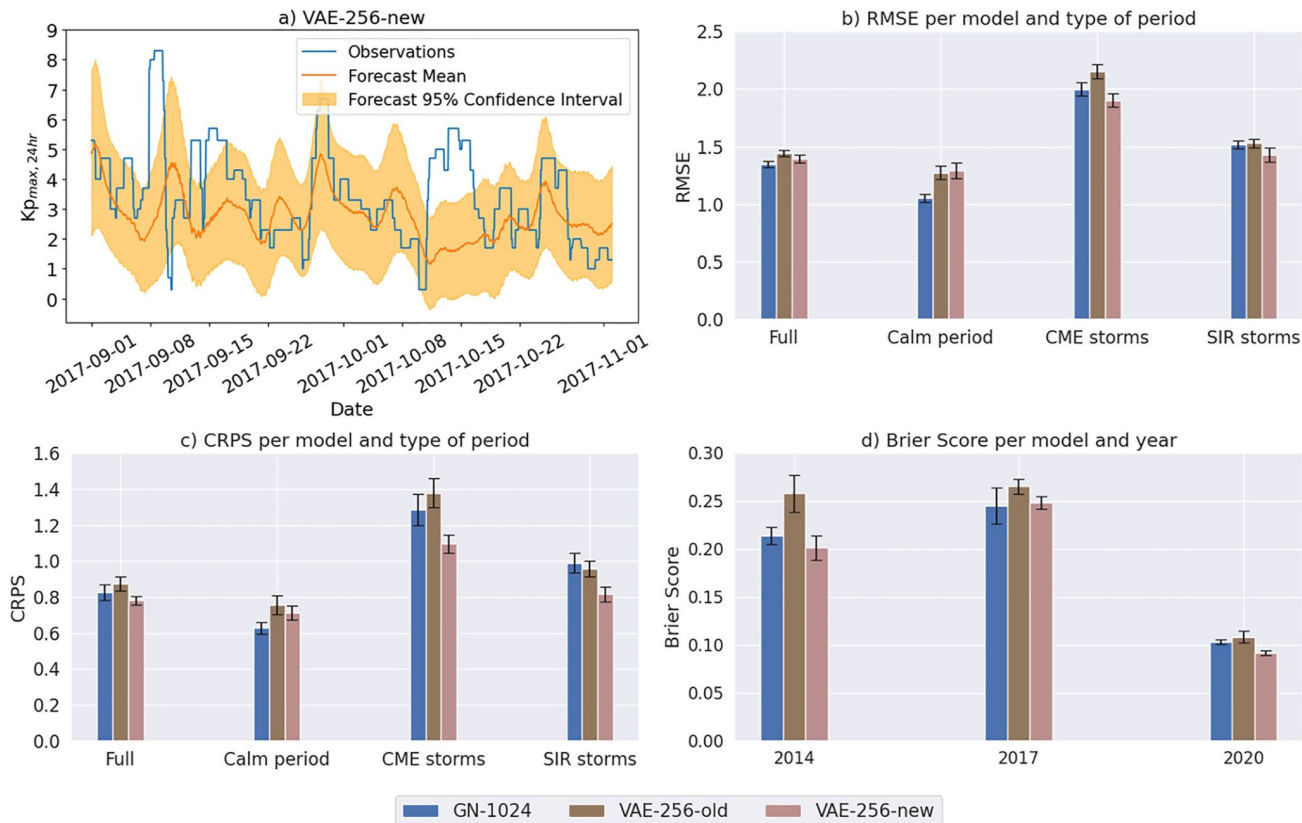


Table 1. Evaluation of Forecasts Made 2-to-7 Days Ahead With SERENADE Over the Full Test Set ($\approx$ 2010–2018) and for Active Periods Alone

Horizon (days)	RMSE		MAE		CRPS		R	
	Full	Active	Full	Active	Full	Active	Full	Active
2	1.35	1.74	1.07	1.41	0.77	1.05	0.27	0.07
3	1.38	1.79	1.09	1.47	0.78	1.09	0.25	0.09
4	1.41	1.79	1.11	1.48	0.80	1.10	0.21	0.11
5	1.43	1.81	1.13	1.52	0.81	1.13	0.21	0.12
6	1.44	1.78	1.15	1.48	0.82	1.10	0.19	0.11
7	1.48	1.82	1.18	1.52	0.85	1.13	0.15	0.09

Tahtouh, M., Bernoux, G., Brunet, A., Standarovski, D., Nguyen, G., & Sicard, A. (2025). Comparison of solar imaging feature extraction methods in the context of space weather prediction with deep learning-based models. Journal of Geophysical Research: Machine Learning and Computation

Exam how different choices of feature extractor affect the performance of the previous model.



Zhelavskaya, R. Vasile, Y. Y. Shprits, C. Stolle, and J. Matzka. Systematic Analysis of Machine Learning and Feature Selection Techniques for Prediction of the Kp Index. Space Weather, 17(10): 1461–1486, October 2019

Input: the solar wind speed (V_{sw}), proton density (n_{Prot}), IMF components B_x , B_y , B_z , and the total IMF magnitude B . We then construct minimum, maximum, and average values of these variables over 1-hr time windows, at L1, from NASA's OMNIWeb data service during the time period of 1998–2017.

Firstly, they compared gradient boosting, neural network, and linear regression and concluded the NN is optimal.

Then, they compared feature selection algorithms and concluded that Fast Function Extraction (FFX) (McConaghy, 2011) is optimal for NN for this task.

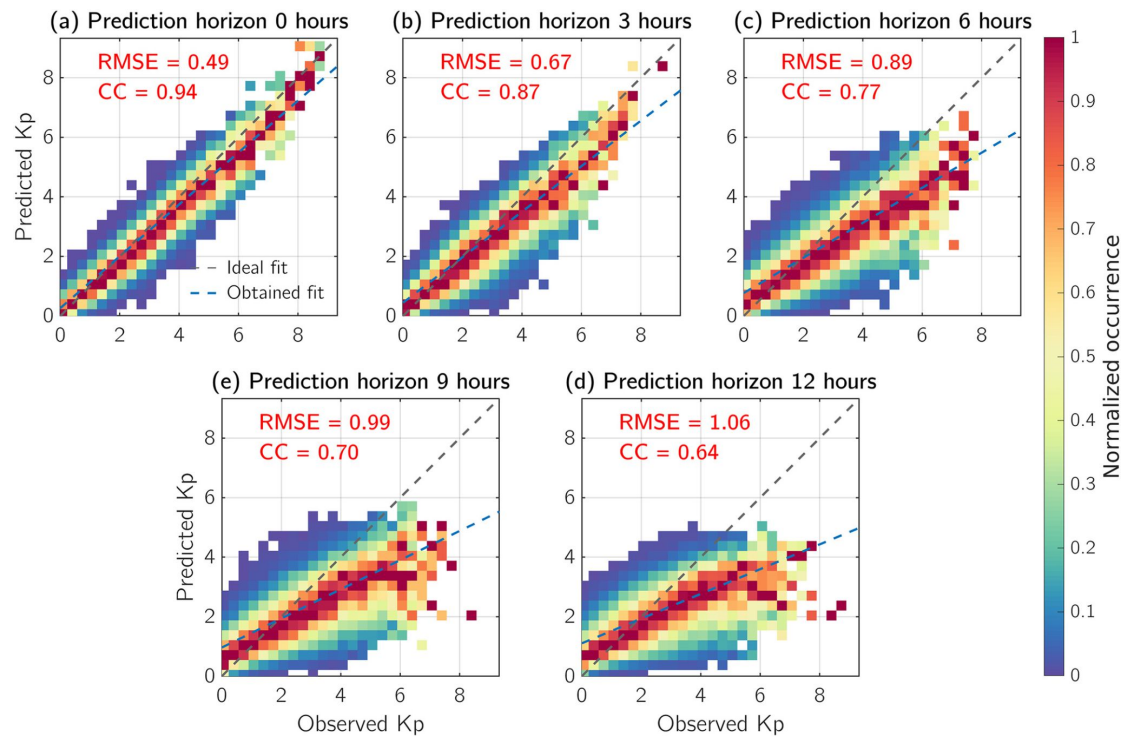


Table 4. Comparison with Existing Kp Predictive Models for Prediction Horizons 0, 3, and 6 hr Ahead

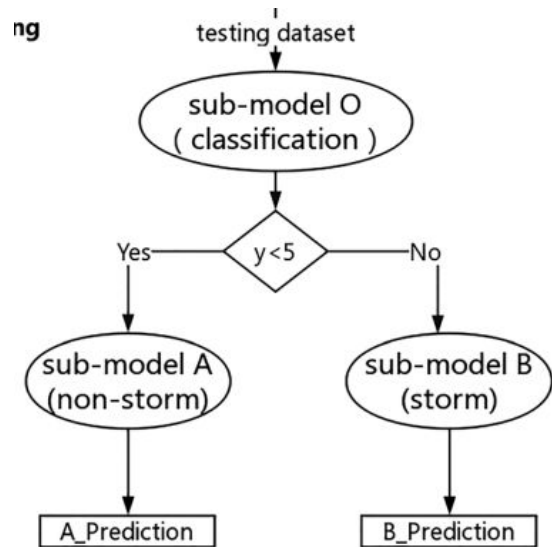
Prediction horizon	Model	RMSE	CC	RMSE NN-FFX model	CC NN-FFX model	Test period
0 (nowcast)	Wintoft et al. (2017)	0.55	0.92	0.51	0.93	2001, 2011
3 hr	Boberg et al. (2000)	0.98	0.77	n/a	n/a	1986–1996
	Bala and Reiff (2012)	0.65	0.86	0.62	0.88	2001/04, 2006/01–2007/12
	Tan et al. (2018)	0.64	0.81	0.65	0.80	2013/12–2014/9
6 hr	Bala and Reiff (2012)	0.85	0.76	0.82	0.78	2001/04, 2006/01–2007/12

Yao Tan, Qinghua Hu, Zhen Wang, and Qiuzhen Zhong. Geomagnetic Index Kp Forecasting With LSTM. Space Weather, 16(4):406–416, April 2018

A training data set (2000.8–2011.10), a validation data set (2012.11–2013.11), and a test data set (2013.12–2014.9)

Table 1
Eleven Candidate Input Parameters in This Research

Parameter	Symbol/Calculating method
The x component of IMF	Bx
The y component of IMF	By
The z component of IMF	Bz
Total magnitude of IMF	B
Flow speed	V
Proton density	D
Boyle index	$BI = 10^{-4}V^2 + 11.7B\sin^3(\theta/2)$
Newell function	$\frac{d\phi_{MP}}{dt} = V^4 B^2 \sin^8(\theta/2)$
Viscous function	$vis = D^{\frac{1}{2}}V^2$
Dynamic pressure	$Dp = 1.6726e - 6 \times DV^2$
Historical Kp	Kp



Submodel	The training algorithm
Model O	LSTM
Model A	LSTM
Model B	Linear regression

Table 4 <i>The Comparison on RMSE, MAE, and CC of Kp Forecasting Among Six Models</i>			
Model	MAE	RMSE	CC
AR model	0.5641	0.7593	0.7334
Linear regression	0.4969	0.6591	0.7997
Lasso regression	0.5091	0.6665	0.7968
SVR	0.4835	0.6522	0.8050
BP neural network	0.4946	0.6596	0.7990
Our model	0.4765	0.6382	0.8147

The Experiment Results Comparing Our Model and Other Existing Models

Model	Input parameter	Prediction time	MAE	RMSE	CC
Boberg (2000)	V, D, Bz	3 hr	0.6070	0.7986	0.6864
Bala (2012)	Newell function, viscous function	3 hr	0.5802	0.7508	0.7368
Our model	Bl, Kp, Dp	3 hr	0.4765	0.6382	0.8147

Peter Wintoft, Magnus Wik, Jürgen Matzka, and Yuri Shprits. Forecasting Kp from solar wind data: input parameter study using 3-hour averages and 3-hour range values. Journal of Space Weather and Space Climate, 7:A29, November 2017

solar wind parameter B , B_y , B_z , n , V , the 3-hour minimum, average, and maximum are calculated

by shifting the timestamps. If a particular solar wind sample i at ACE has timestamp and data (t_i, x_i) the propagated sample will have

$$(s_i, x_i) = (t_i + \tau_i, x_i), \quad (1)$$

where $\tau_i = r/v_i$, r is the distance from ACE to 10 Earth radii ($10R_E$) upstream from Earth and v is the speed. We
Datasets: (1998, 1999, 2002–2005, 2007–2010, 2013–2015), (2000, 2006, 2012), and (2001, 2011)

Feed-forward neural network with one hidden layer

Forecasting horizon is r/v (dynamic)

